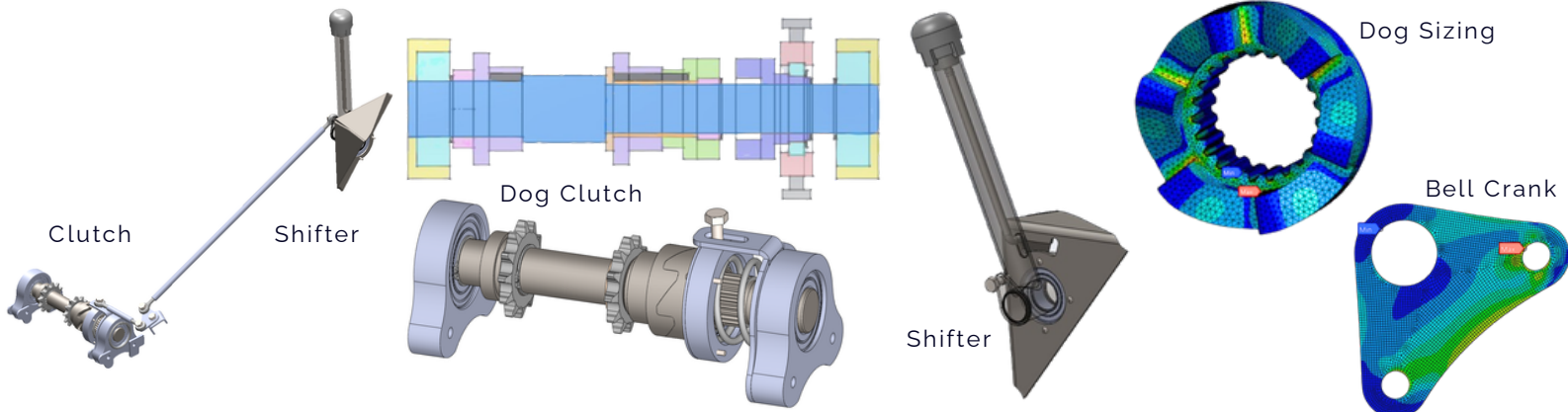


INTERMEDIATE JACKSHAFT AND DOG CLUTCH



What?

- Converted fixed jackshaft to **shiftable driveline**, requiring engagement during rotation
- Initial geometry showed stress concentrations at spline root and hub interface, **exceeding material limits** under peak loading
- Actuation needed to overcome spring forces (**150–200 N range**) while avoiding linkage instability

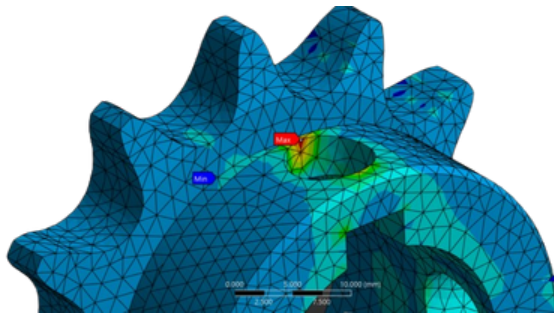
How?

- Modeled drivetrain inertia and compliance to estimate peak impact torque, sizing dog teeth and hub accordingly
- Increased effective torque radius to **reduce shear stress** at the interface and distribute load
- Designed linkage using buckling and friction constraints, selecting geometry that maintained **FoS >1.5**

Results

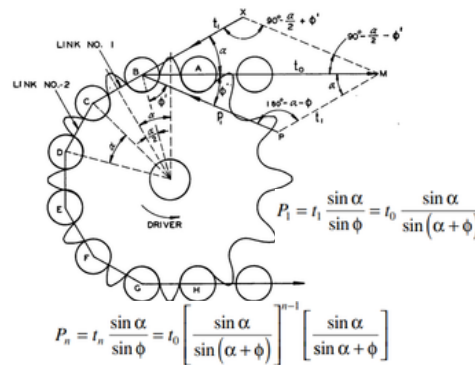
- Reduced peak stress in critical regions by **~40%** in **ANSYS** through geometry resizing
- Maintained stable actuation under full spring preload, with no **buckling** or **excessive deflection** observed
- Final design met target loading conditions analytically, ready for physical validation

SPROCKET AND DRIVETRAIN ANALYSIS



Sprocket Failure at Set Screw

Tooth Load Distribution



What?

- Sprocket repeatedly **failed at set screw interface** despite tooth loading being primary torque path
- Original drivetrain **estimates overstated available power** (14HP assumed vs 9.5HP actual), skewing load assumptions

How?

- Simulated** over **20 cases**, varying applied loads, contact locations, and fixtures
- Applied **tooth load distribution** to increase simulation accuracy
- Corrected previous calculations, increasing component sizing and type to compensate.
- Modeled engagement event using **lumped torsional modelling** to estimate peak torque during clutch engagement

Results

- FEA predicted stress concentration at set screw **matching real fracture location**, validating failure mechanism
- Set screw preload increased local stress by **2x** compared to torque loading alone
- Identified baseline sprocket operating near **material yield** (~280 MPa), explaining repeated failure under shock

Powertrain Torsional Impact Model:

Reduced inertia:	$J_{eq} = \frac{J_1 J_2}{J_1 + J_2}$
Initial mismatch energy	$E = \frac{1}{2} \frac{J_{eq}}{g_c} \Delta \omega^2$
Chain axial stiffness:	$k_{axial} = \frac{EA}{L}$
Sprocket torsional stiffness at pitch radius	$k_t = k_{axial} r_s^2$
Peak torque from energy storage:	$T_{max} = \sqrt{2Ek_t}$
Tangential tooth force at clutch avg radius:	$F_{total} = \frac{T_{max}}{r_m}$
Load per tooth:	$F_{tooth} = \frac{F_{total}}{n_{engaged}}$

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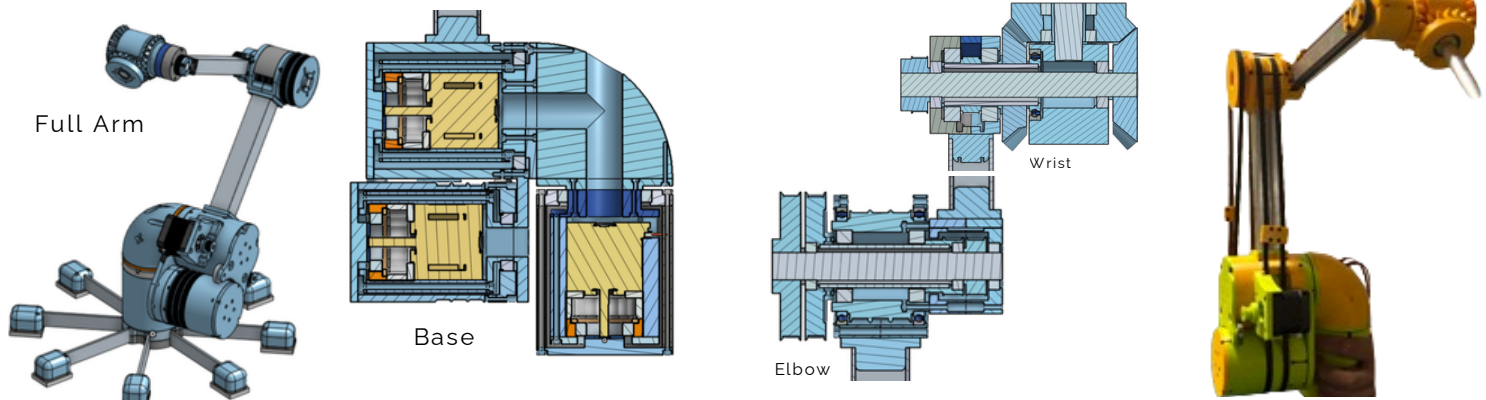
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LOW COST ROBOTIC ARM



What?

- Desired a fast, low cost robotic arm targeting **< \$100**
- Off the shelf layouts were either too bulky or required **expensive hardware** (actuators, structure, bearings)
- Main challenge became managing joint loads and stiffness with **lightweight, low cost** components

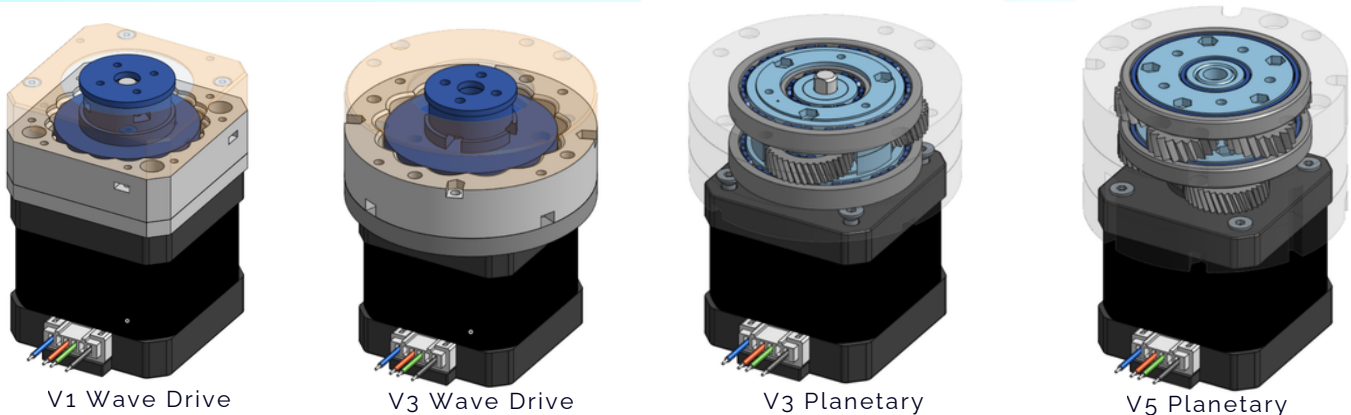
How?

- Concentrated mass at the base, using heavy actuators to **passively counterweigh loading**
- Developed custom low friction, low play, any size bearings to reduce cost and further compact and lightweight the design
- Designed structure around **bending moments**, reinforcing high stress regions

Results

- Reduced estimated joint loading by **35%**, enabling smaller actuators and lighter link geometry
- Cut hardware cost by **80%** compared to typical hobby arms, while maintaining comparable reach and functionality
- Achieved a **compact, modular arm** platform ready for interchangeable end-effectors and control integration

LOW COST COMPACT 3D PRINTED GEARBOX



What?

- Needed a **sub \$10**, fully 3D printed actuator capable of **>1.6Nm in a <60 mm envelope**
- Early compact wave style prototype failed under load (**0.9 Nm limit**) and introduced severe vibration
- Key constraint became achieving reliable torque density with low stiffness printed materials

How?

- Transitioned to a 2 stage planetary, distributing ratio across stages to **isolate high torque** to reinforced geometry
- Tuned gear modules to balance compactness with tooth strength
- Built equation driven **parametric models** to rapidly change diameters, ratios, and tooth counts across **10+ configurations**

Results

- Torque capacity increased from 0.9 Nm → **2 Nm** under equivalent geometry constraints
- Eliminated vibration observed in wave drive through rigid gear engagement and **improved load paths**
- Converged on compact **17:1 drivetrain** with stable meshing behavior

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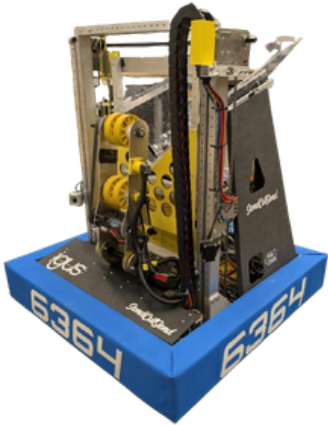
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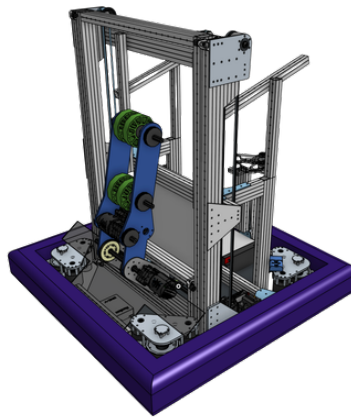
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FIRST ROBOTICS COMPETITION



What?

- Needed **fast cycling and reliable robot**
- Early concepts either optimized one subsystem or another, but integration issues killed consistency
- Core challenge became building a system where **drive, intake, and scoring all worked together** without conflict



How?

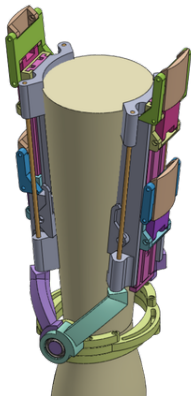
- Designed around a **low, stable base** with clean packaging to support aggressive driving and consistent alignment
- Developed an **intake that corrected game elements**, and folded up, allowing for endgame mechanisms
- Built a **cascade elevator** with controlled motion and geometry to maintain stability while extending and scoring



Results

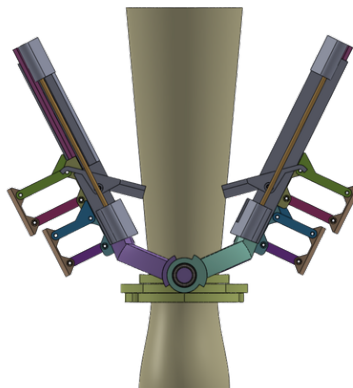
- Achieved fast, repeatable cycles with **minimal driver correction** required
- Was able to support a large intake and hang during endgame, **increasing average score by 30%**
- Delivered a well integrated system where subsystems reinforced each other instead of competing

EXPANDABLE ARMOUR MECHANISM



What?

- Desired a **compact wearable mechanism** that could dramatically expand on deployment
- Early concepts either looked cool but were too bulky, or were compact but jammed / **deployed unreliably**
- Core challenge became balancing visual impact with mechanical simplicity and repeatability



How?

- Iterated through 3 design build test cycles, refining **linkage geometry** and **sliding interfaces** to control motion paths
- Used **parametric CAD** to tune ergonomics and size, eliminating binding and ensuring smooth, guided expansion
- Applied **DFMA** to merge parts and simplify joints, reducing stack up error and easing assembly/testing



Results

- Cut thickness from ~40 mm → 21 mm (**47.5%**) while maintaining full expansion behavior
- Achieved smooth, repeatable deployment across iterations with **no major jamming or misalignment**
- Final prototype delivered a compact, wearable form with a **high impact expansion sequence**, validating the design approach

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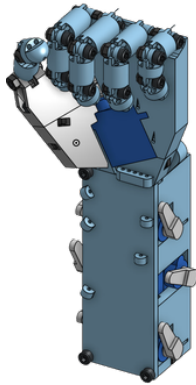
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ROBOTIC HAND



What?

- Needed a compact robotic hand that could move naturally without complex linkages or bulky hardware
- Core challenge became achieving human like motion using **simple, low cost actuation**

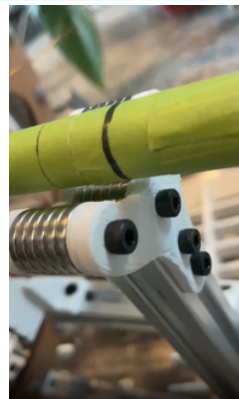
How?

- Used a tendon driven system with routed lines and elastic return to mimic muscle behavior
- Iterated finger geometry to balance thin sections for flexibility with reinforcement in **high stress regions**
- Tuned tendon paths and anchor points to **improve motion consistency and reduce friction**

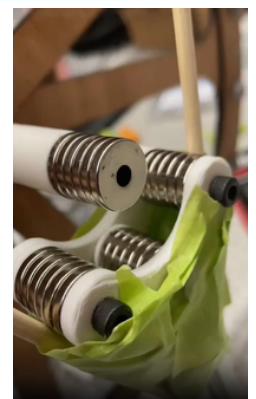
Results

- Achieved smooth, coordinated finger motion with a **minimal servo setup**
- Increased finger durability and output force significantly.
- Delivered a fully functional, **lifelike prototype** suitable for demos despite tight constraints

MAGLEV HYDRO GENERATOR



Levitation!!



What?

- Tried to build a generator that could eliminate mechanical losses, targeting higher efficiency through **frictionless rotation**
- Concept pushed into a conflict where levitation forces and power generation fields interfered with each other
- Core challenge became whether a system could both **levitate and generate efficiently**

How?

- Built a **magnetically supported rotor** using stacked permanent magnets to stabilize and lift the rotating mass
- Designed **custom coils and a flywheel** to smooth rotation and extract usable electrical output
- Iterated **magnet layout and spacing** to balance stability vs. electromagnetic interaction

Results

- Achieved **stable magnetic support** with continuous rotation, validating the levitation concept
- Identified major efficiency losses due to field **interference between levitation and generation systems**
- Demonstrated that combining both functions introduced **tradeoffs**, shaping a clear direction for separating systems in future designs